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Paper Code : 21304

F–404

B. C. A. (First Semester)
EXAMINATION, 2017-18
(New Course)

Paper No. BCA—104—N

MATHEMATICS—I

Time : Three Hours] [*Maximum Marks : 70*

Note : Attempt any *five* questions. All questions carry equal marks. Symbols used have their usual meanings.

1. (a) Find the right hand and left hand limits of $\lim_{x \to 0} \frac{e^{1/x} - 1}{e^{1/x} + 1}$.

   (b) Show that the function f defined as :

$$f(x) = \begin{cases} 0 & : \ x = 0 \\ \frac{1}{2} - x & : \ 0 < x < \frac{1}{2} \\ \frac{1}{2} & : \ x = \frac{1}{2} \\ \frac{3}{2} - x & : \ \frac{1}{2} < x < 1 \\ 1 & : \ x = 1 \end{cases}$$

   has three points of discontinuity which you are required to find.

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2. (a) If : $y = \sin mx + \cos mx$

   Prove that :

$$y_n = m^n \left[1 + (-1)^n \sin 2mx\right]^{1/2}$$

   (b) If : $y = e^{a \cos^{-1} x}$

   prove that :

$$(1 - x^2) y_{n+2} - (2n + 1) x y_{n+1} - (n^2 + a^2) y_n = 0$$

3. (a) If : $u = \tan^{-1} \frac{xy}{\sqrt{1 + x^2 + y^2}}$

   show that :

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{1}{(1 + x^2 + y^2)^{3/2}}$$

   (b) State and prove Euler's theorem.

4. The roots of the equation in λ :

$$(\lambda - x)^3 + (\lambda - y)^3 + (\lambda - z)^3 = 0$$

   are u, v, w.

   Prove that :

$$\frac{\partial(u, v, w)}{\partial(x, y, z)} = -2 \frac{(x - y)(y - z)(z - x)}{(u - v)(v - w)(w - u)}$$

5. (a) If : $I_n = \int_0^{\pi/4} \tan^n x \, dx$

   prove that :

$$I_{n+1} + I_{n-1} = \frac{1}{n}$$

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(b) Evaluate $\iint x^2 y^2 dx\, dy$ over that region $x^2 + y^2 \leq 1$.

6. (a) Find the whole area of the curve :

$$a^2 y^2 = x^3 (2a - x)$$

(b) Show that the volume of a sphere of radius a is $\frac{4}{3}\pi a^3$.

7. (a) Show that :

$$\hat{r} \times d\hat{r} = \frac{\hat{r} \times d\hat{r}}{r^2}$$

(b) Prove that :

$$\text{div grad } r^n = n(n+1) r^{n-2}$$

8. (a) Solve :

$$\frac{dy}{dx} + 2y \tan x = \sin x$$

given that $y = 0$ when $x = \frac{\pi}{3}$.

(b) Solve :

$$(D^2 - 2D + 1) y = x e^x \sin x$$

9. (a) Discuss the properties of parabola and hyperbola.

(b) Trace the ellipse :

$$3x^2 + 4y^2 = 12$$

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